

Compactoid Mackey convergence in metrizable locally convex spaces

Solution packet for arXiv:1511.09045

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Sharp resolution. Over every nontrivially valued field, the source conjecture is true in the stronger class of metrizable locally convex spaces: topological convergence, Mackey convergence for the von Neumann bornology, and Mackey convergence for the compactoid bornology are equivalent. Over a trivially valued field the unrestricted statement is false, already for the Fréchet space $k^{\mathbb{N}}$.

Source conjecture

For a locally convex k -space E , let E^b denote its von Neumann bornology and $\text{Cpt}(E)$ its compactoid bornology. A sequence (x_n) converges Mackey-bornologically to zero if there is a bounded set B such that, for every $\lambda \in k^\times$, eventually $x_n \in \lambda B$.

After proving the following equivalence for nuclear Fréchet spaces, the source conjectures that nuclearity is unnecessary.

Lemma 3.68. *Let E be a nuclear Fréchet space and $\{x_n\}$ a sequence of elements of E . Then, the following are equivalent*

- (1) $\{x_n\}$ converges topologically to 0;
- (2) $\{x_n\}$ converges bornologically to 0 for the bornology of E^b ;
- (3) $\{x_n\}$ converges bornologically to 0 for the bornology of $\text{Cpt}(E)$.

Proof. This result is proven for Archimedean base fields in [25] Corollary 3.8. For the general case: For the equivalence of (1) and (2) we can refer to [3], Proposition 1.17. Finally, conditions (2) and (3) are equivalent by Lemma 3.67. $\square$

Remark 3.69. We conjecture that Lemma 3.68 holds for all Fréchet spaces, without the nuclearity hypothesis. Indeed the proof given in [25] for Archimedean base fields, does not use the nuclearity hypothesis but it easily adapts to the non-Archimedean case only when the base field is locally compact. Notice that the only missing implication for a general non-Archimedean base field is (3) implies (2) or (3) implies (1).

Corollary 3.70. *Let $f : E \rightarrow E$ be a morphism of nuclear Fréchet spaces. Then, the*

There is a harmless reversal in the last sentence of the source remark: compactoid sets are always von Neumann bounded, so (3) $\Rightarrow$ (2) is immediate. The substantive direction is (1) $\Rightarrow$ (3), proved below.

Slow rescaling of a null sequence

Lemma 1. *Let k be nontrivially valued and let E be a metrizable locally convex k -space. If $x_n \rightarrow 0$ topologically, then there are $a_n \in k^\times$ such that $|a_n| \rightarrow \infty$ and $a_n x_n \rightarrow 0$ topologically.*

Proof. Choose $\pi \in k$ with $0 < |\pi| < 1$, and choose an increasing defining sequence of seminorms $p_1 \leq p_2 \leq \dots$ for E . Recursively choose $N_1 < N_2 < \dots$ so that

$$p_j(x_n) \leq |\pi|^{2^r} \quad (n \geq N_r, 1 \leq j \leq r).$$

For $N_r \leq n < N_{r+1}$ set $a_n = \pi^{-r}$ (and define the finitely many earlier a_n arbitrarily). Then $|a_n| \rightarrow \infty$. If $r \geq j$ and $N_r \leq n < N_{r+1}$, then

$$p_j(a_n x_n) \leq |\pi|^{-r} |\pi|^{2r} = |\pi|^r \rightarrow 0.$$

Thus $a_n x_n \rightarrow 0$ in every defining seminorm. $\square$

Affirmative theorem for nontrivial valuations

Theorem 1. *Let k be a nontrivially valued field and E a metrizable locally convex k -space. For every sequence $(x_n) \subset E$, the following are equivalent:*

- (1) $x_n \rightarrow 0$ topologically;
- (2) $x_n \rightarrow 0$ Mackey-bornologically in E^b ;
- (3) $x_n \rightarrow 0$ Mackey-bornologically in $\text{Cpt}(E)$.

In particular the source conjecture holds for every Fréchet space over a nontrivially valued field, without nuclearity or local compactness of the field.

Proof. Assume (1). Apply Lemma 1 and put $y_n = a_n x_n$. The set

$$K = \{0, y_1, y_2, \dots\}$$

is compact: any open set containing zero contains all but finitely many y_n . Hence K is precompact and therefore compactoid. Its absolutely convex hull $B = \Gamma(K)$ is also compactoid. Indeed, if U is an absolutely convex zero-neighborhood and $K \subset U + F$ for a finite F , then

$$\Gamma(K) \subset U + \Gamma(F).$$

Now $x_n = a_n^{-1} y_n$ and $|a_n^{-1}| \rightarrow 0$. For any fixed $\lambda \in k^\times$, eventually $|a_n^{-1}/\lambda| \leq 1$; absolute convexity gives $x_n \in \lambda B$. Thus (3) holds.

Every compactoid set is von Neumann bounded, so (3) $\Rightarrow$ (2). Finally, suppose (2) is witnessed by a bounded set B . Given a zero-neighborhood U , boundedness gives a scalar $c \in k^\times$ with $B \subset cU$. Mackey convergence, applied to c^{-1} , yields eventually $x_n \in c^{-1}B \subset U$. Hence (2) $\Rightarrow$ (1). $\square$

The trivially valued obstruction

Theorem 2. *Let k be a nonzero trivially valued field. The conjectured equivalence fails for the Fréchet space $E = k^\mathbb{N}$ with its product topology.*

Proof. Give each copy of k its norm topology; because the valuation is trivial, it is discrete. The product E is complete and metrizable. Its standard unit vectors e_n converge coordinatewise, hence topologically, to zero.

We claim that every von Neumann bounded subset of E is contained in $\{0\}$. If a bounded set B contained a nonzero x , choose a coordinate j with $x_j \neq 0$ and the absolutely convex open zero-neighborhood

$$U_j = \{z \in E : z_j = 0\}.$$

For every $\lambda \in k^\times$, triviality of the valuation gives $\lambda U_j = U_j$, so no scalar multiple of U_j contains x . This contradicts boundedness. Consequently a Mackey-null sequence in E^b (and therefore in $\text{Cpt}(E)$) must be eventually zero. The sequence (e_n) is not eventually zero, so (1) does not imply (2) or (3). $\square$

Audit notes

- Completeness is not used in the positive theorem; metrizability is exactly what permits the diagonal slow rescaling.
- The proof does not identify the two bornologies. They can differ greatly; only their convergent sequences agree.
- The absolutely convex hull step is valid because $U + \Gamma(F)$ is absolutely convex and contains K .
- Local compactness and spherical completeness of k are not used.
- Nontriviality of the valuation is essential, as Theorem 2 shows.

Source

F. Bambozzi, *Stein domains in Banach algebraic geometry*, arXiv:1511.09045 (source Remark 3.69 in the current PDF).